

Project Progress Report: Fusion-Baukasten

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Overview

This report outlines the technical foundation and infrastructure work completed to date for the **Fusion-Baukasten** project. The focus of this initial phase was to translate the design prototypes and business logic into a robust, secure, and scalable software architecture before beginning the visual frontend development.

Completed Milestones

1. Repository Setup & Version Control

- Established a professional GitHub repository to track all code changes.
- Cleaned up the directory structure by separating heavy design files (like Figma exports and large PDFs) from the core source code, ensuring the repository remains fast and maintainable.
- Set up a secure .gitignore to prevent sensitive credentials and environment variables from being leaked.

2. Database Architecture (PostgreSQL)

- Designed and deployed the complete database schema required for the application.
- Successfully created 14 distinct database tables covering all core domains:
 - **User Management** (Users, Roles)
 - **Project Management** (Projects, Teams)
 - **Dialogue System** (Flows, Blocks, Answers)
 - **Collaboration** (Tasks, Activity Logs)
 - **Knowledge Base** (AI Context, Vector Storage)
- Configured automated database migrations (Alembic) so future updates to the data structure can be rolled out smoothly without data loss.

3. Backend API & Security (FastAPI)

- Initialized the backend server using FastAPI (Python), chosen for its high performance and modern architecture.
- Implemented a complete Authentication System.
- Built secure user login and registration endpoints using industry-standard JWT (JSON Web Tokens) and bcrypt password hashing.

4. Frontend Foundation (Next.js)

- Bootstrapped the frontend application using Next.js 15 and React.
- Configured the new TailwindCSS (v4) design system to exactly match the brand colors and typography provided in the design prototypes.
- Transitioned from a "build the whole UI at once" approach to a more stable "Modular Feature-by-Feature" approach. This ensures every piece of the application is fully functional (frontend +

backend) before moving to the next.

Next Steps

With the invisible "plumbing" (database, server, and security) fully operational, the immediate next phase is **Feature 1: Dashboard & Authentication UI**.

I will now begin connecting the visual frontend screens directly to the backend systems we just built, starting with the user login experience and the main project dashboard.